

DevOps Lab Program- 5

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Project Structure:

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-5$ tree
.
├── docker-compose.yml
└── Dockerfile

1 directory, 2 files
```

Steps for execution:

Step 1: Ensure docker service is running

```

1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1-24.04.1
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-$ docker compose version
Docker Compose version v5.0.1
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-$ sudo systemctl status docker
[sudo] password for 1rv24mc005_aditi:
Sorry, try again.
[sudo] password for 1rv24mc005_aditi:
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-01-21 22:32:02 IST; 3h 51min ago
     TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
    Main PID: 2102 (dockerd)
      Tasks: 12
     Memory: 428.0M (peak: 485.8M)
        CPU: 38.596s
       CGroup: /system.slice/docker.service
               └─2102 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Jan 21 22:32:02 aditi-Inspiron-3584 dockerd[2102]: time="2026-01-21T22:32:02.209156458+05:30" level=info msg="Initializing buildkit"
Jan 21 22:32:02 aditi-Inspiron-3584 dockerd[2102]: time="2026-01-21T22:32:02.222133193+05:30" level=warning msg="CDI setup error /etc/cdi: failed to mon-
Jan 21 22:32:02 aditi-Inspiron-3584 dockerd[2102]: time="2026-01-21T22:32:02.222672418+05:30" level=warning msg="CDI setup error /var/run/cdi: failed to
Jan 21 22:32:02 aditi-Inspiron-3584 dockerd[2102]: time="2026-01-21T22:32:02.324795798+05:30" level=info msg="Completed buildkit initialization"
Jan 21 22:32:02 aditi-Inspiron-3584 dockerd[2102]: time="2026-01-21T22:32:02.347977926+05:30" level=info msg="Daemon has completed initialization"
Jan 21 22:32:02 aditi-Inspiron-3584 dockerd[2102]: time="2026-01-21T22:32:02.348116102+05:30" level=info msg="API listen on /run/docker.sock"
Jan 21 22:32:02 aditi-Inspiron-3584 systemd[1]: Started docker.service - Docker Application Container Engine.
Jan 22 00:57:01 aditi-Inspiron-3584 dockerd[2102]: time="2026-01-22T00:57:01.214086769+05:30" level=info msg="Container failed to exit within 10s of sig-
Jan 22 00:57:01 aditi-Inspiron-3584 dockerd[2102]: time="2026-01-22T00:57:01.300231739+05:30" level=info msg="ignoring event" container=79ecac2519e2aa0b-
Jan 22 01:12:05 aditi-Inspiron-3584 dockerd[2102]: time="2026-01-22T01:12:05.755662004+05:30" level=info msg="ignoring event" container=a7fedac868ead06b-
[lines 1-22/22 (END)]

```

Step 2: Create a **Dockerfile**:

docker run -p 8080:8080 -p 50000:50000 jenkins-img
(or alternatively, *docker compose up*)

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-5$ docker run -p 8080:8080 -p 50000:50000 jenkins-img
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2026-01-21 21:09:47.612+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2026-01-21 21:09:49.384+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-01-21 21:09:49.543+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.1.5; built: 2025-12-03T22:18:24.732Z; git: 4905d09d7b68
01e792d7b73946cf7b66e4be25d6; jvm 21.0.9+10-LTS
2026-01-21 21:09:50.714+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.
ee9.jsp.JettyJspServlet
2026-01-21 21:09:50.892+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-01-21 21:09:51.931+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkins_home found at: EnvVars.masterE
nvVars.get("JENKINS_HOME")
2026-01-21 21:09:52.282+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@3681037{Jenkins
v2.541.1,,/b=file:///var/jenkins_home/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@2459319c{STARTED}}
2026-01-21 21:09:52.324+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@2aa6311a{HTTP/1.1, (http/1.1)}{0.0.0
.0:8080}
2026-01-21 21:09:52.327+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@78525ef9{STARTING}[12.1.5,sto=0] @2789ms
2026-01-21 21:09:52.330+0000 [id=34] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-01-21 21:09:52.613+0000 [id=33] INFO jenkins.model.Jenkins#<init>: Starting version 2.541.1
2026-01-21 21:09:52.841+0000 [id=42] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-01-21 21:09:52.878+0000 [id=41] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-01-21 21:09:55.280+0000 [id=45] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-01-21 21:09:55.290+0000 [id=43] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-01-21 21:09:55.293+0000 [id=45] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-01-21 21:09:55.815+0000 [id=44] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-01-21 21:09:55.819+0000 [id=42] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-01-21 21:09:55.820+0000 [id=42] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-01-21 21:09:55.828+0000 [id=44] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-01-21 21:09:55.977+0000 [id=61] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-01-21 21:09:57.393+0000 [id=40] INFO jenkins.install.SetupWizard#init:
```

Step 5: Verify the expected output by logging into Jenkins at <http://localhost:8080>

localhost:8080

Getting Started

Unlock Jenkins

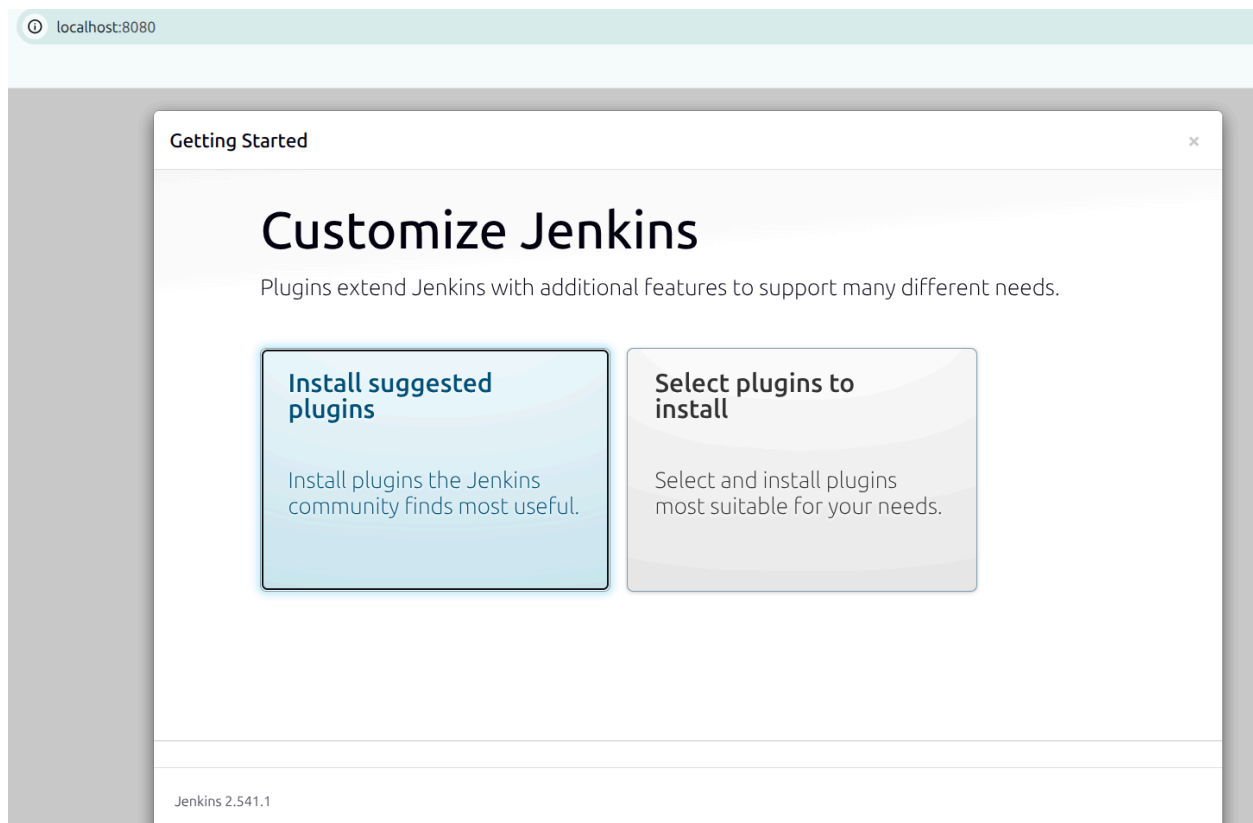
To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

```
/var/jenkins_home/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Continue



Step 6: Stop and remove the Jenkins image

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-5$ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
program-5-jenkins   latest          eef5e7e6bfa8   9 minutes ago   924MB
jenkins-img         latest          ff294e09cf40   9 minutes ago   924MB
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-5$ docker compose down
WARN[0000] /home/1rv24mc005_aditi/dockerFolder/program-5/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to
avoid potential confusion
[+] down 2/2
✓ Container jenkins-ci      Removed      0.8s
✓ Network program-5_default Removed      0.2s

1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-5$ docker rmi program-5-jenkins:latest jenkins-img:latest
Untagged: program-5-jenkins:latest
Deleted: sha256:eef5e7e6bfa81934a32aed08ba0d7963fd36e586580eaf4ed2ccf972f703e8ba
-
Untagged: jenkins-img:latest
Deleted: sha256:ff294e09cf408039a1d6b71549bd673b7f7f8ebc36eea096793d2e2e919fc9ed
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-5$ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
```